



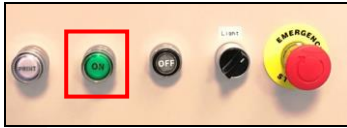
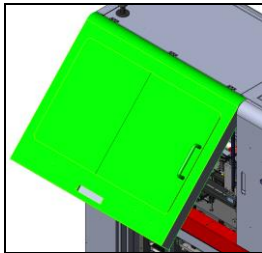
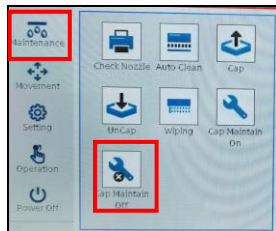
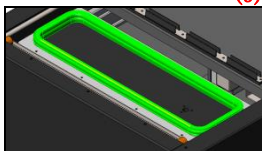
Product: MagicSpin Pro			From: Technical Writing Department	
Version	Date	Revision	Written by	Reviewed by
01	2024-11-6	Creation	Myra	

Maintenance Guide

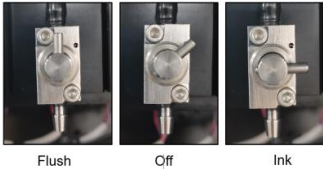
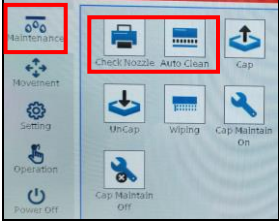
The maintenance in this document is for the end user. Maintain the printer according to the following table. If you have any issues or doubts, contact your technical support.

NOTE: The maintenance schedule is based on AMICA ink and an 8-hour workday. If your printer works with a higher load or the printing environment is dusty, maintain the printer more frequently.

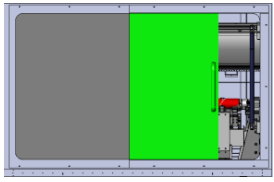
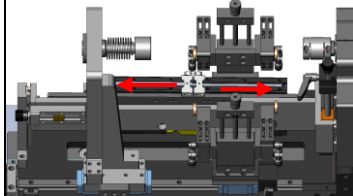
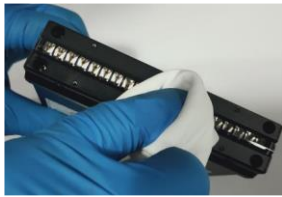
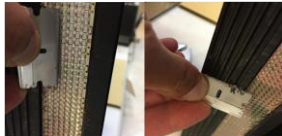
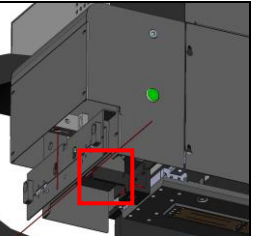
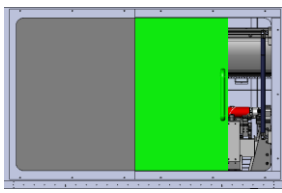
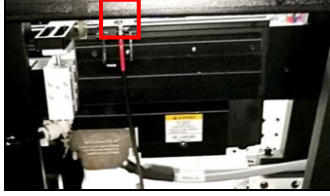
DAILY MAINTENANCE

Items	Tools	Procedures	Pictures
Clean the head plate and the PWC	Cleanroom wiper Latex powder-free gloves Flushing fluid	- Power on the printer. - Move the carriage to the left end of the carriage beam by tapping the “Left Arrow” button on the touch-screen LCD. - Open the flip-up door. - Fold the Cleanroom wiper and spray the flushing fluid on it. - Wipe off the ink on the edges and undersurface of the plate with the wipers. NOTE: Do not touch the nozzle plate of the print head while cleaning. - Close the flip-up door and open the right door. - Clean the wiping blades with a new Cleanroom wiper soaked with flushing fluid. - On the touch-screen LCD, tap “Maintenance > Cap Maintain On” to reveal the sealing rubber strip. - Use the Cleanroom wiper and flushing fluid to clean the sealing rubber strip. - Close the right door. - On the touch-screen LCD, tap “Maintenance > Cap Maintain Off” to conceal the sealing rubber strip.	 (1)  (2)  (3)  (4)  (5)  (6)  (8)  (11)  (9)

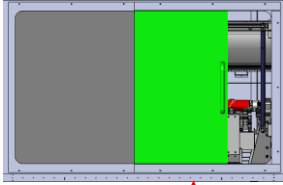
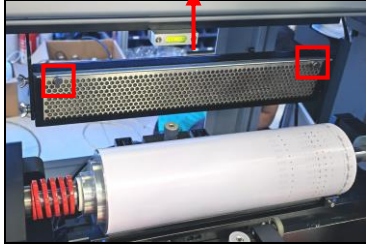
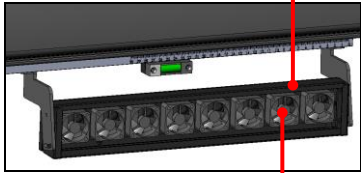


Items	Tools	Procedures	Pictures
Inspect head nozzles	Magnifier (10X or higher)	- Switch the manual 3-way valve to position “INK” with the wrench. - Perform an auto clean: on the Touch-screen LCD, select “Maintenance > Auto Clean”. - Print the check nozzle chart to inspect the head nozzle status. Write down the date, time, and your name on the chart. - Clear the nozzles by purging ink or flushing fluid if necessary.	 

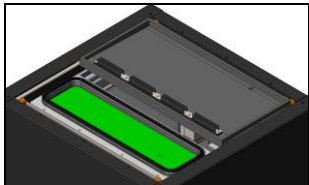
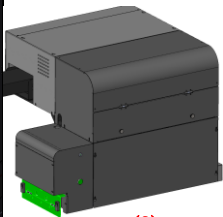
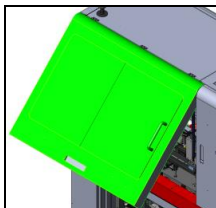
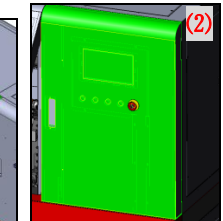
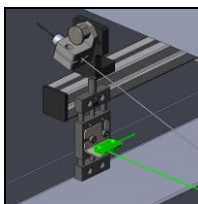
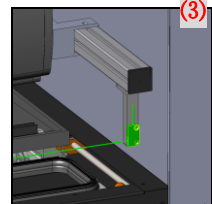
■ EVERY TWO DAYS

Items	Tools	Procedures	Pictures
Clean the emitting window	Cleanroom wiper Latex powder-free gloves IPA Razor blade	1. Power off the printer. 2. Open the sliding door. 3. Unload the taper/cylinder from the printer. 4. Slide the UV light shield to reveal the emitting window. 5. Use the Cleanroom wipers pre-moistened with IPA to remove dust, debris, or cured ink on the emitting window. Clean the emitting window repeatedly if necessary. ⚠ WARNING: The Cleanroom wiper cannot be too wet to prevent IPA from dripping into the spin tool module or other live parts. 6. If the ink, dust, or debris on the emitting window becomes difficult to remove, use a razor blade to remove them. 7. If needed, use the dry Cleanroom wipers to wipe the light source. 8. Close the sliding door.	 (2)  (4)  (5)  (6) *The picture is for reference only.
Clean the UV light sensor	Latex powder-free gloves Cleanroom wiper Industrial alcohol	1. Power off the printer. 2. Open the right door. 3. Wipe the emitting window of the sensor with the Cleanroom wiper moistened with industrial alcohol.	 (2)  (3)
Clean the UV light shield sensor	Latex powder-free gloves Cleanroom wiper Industrial alcohol	1. Power off the printer. 2. Open the sliding door. 3. Unload the taper/cylinder from the printer. 4. Wipe the emitting window of the sensor with the Cleanroom wiper moistened with industrial alcohol. 📝 NOTE: The Cleanroom wiper cannot be too wet to prevent industrial alcohol from dripping onto the UV lamp and causing short circuits. 5. Close the sliding door.	 (2)  (3)

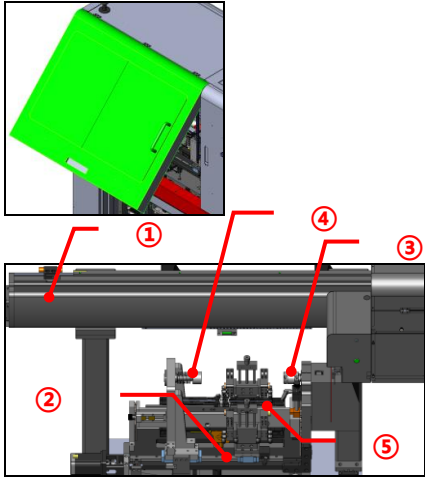
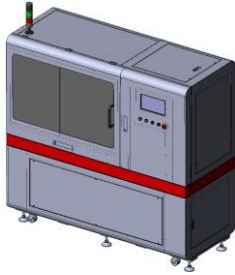
■ EVERY FIVE DAYS

Items	Tools	Procedures	Pictures
Clean the Ink Mesh	Latex powder-free gloves	1. Power off the printer.	 (2)
	Cleanroom wiper	2. Open the sliding door.	
	Industrial alcohol	3. Hold the handle and pull out the ink mesh.	 (3)
	Rubber gloves	4. Soak the ink mesh in alcohol for at least 30 minutes, shake the ink mesh, and remove the waste ink on it with Cleanroom wipers.	
	Mask	5. Put the ink mesh in the clean alcohol again; shake it and remove the remaining waste ink with Cleanroom wipers.	 (4)
	Container	Repeat this step several times until the alcohol is clear.	
	Protective goggles	6. Take out the ink mesh, drain it, and wipe it with dry Cleanroom wipers. And then air-dry or blow-dry it.	 (7) Fan Cover Ink Mist Fans
	Air gun	⚠ WARNING: Never install the ink mesh when it is wet. Otherwise, alcohol may get on the live parts and cause printer damage.	
		7. Wipe off the waste ink on the ink mist fans and the fan covers with Cleanroom wipers and industrial alcohol. ⚠ WARNING: The Cleanroom wipers cannot be too wet. Otherwise, alcohol may get on the live parts and cause printer damage.	
		8. Install the cleaned and dried ink mesh to the ink mist collector.	
		9. Close the sliding door.	

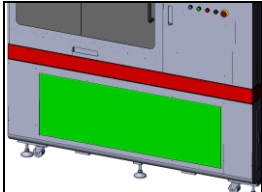
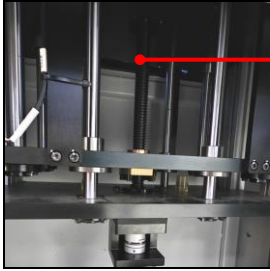
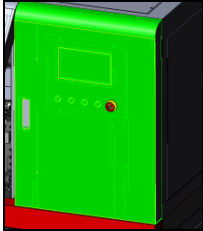
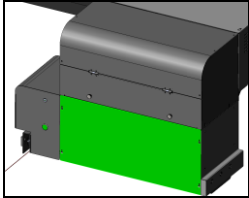
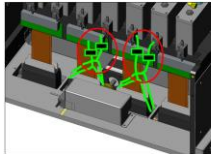
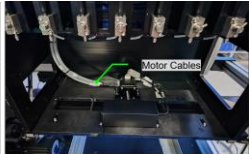
■ WEEKLY MAINTENANCE

Items	Tools	Procedures	Pictures
Clean the wiping tray	Latex powder-free gloves Cleanroom wiper Flushing fluid	1. To reveal the wiping tray, tap the “Left Arrow” button from the “Home” page on the touch-screen LCD. 2. Power off the printer. 3. Open the right door. 4. Clean the tray with Cleanroom wipers and the flushing fluid. 5. Dry the tray with new and dry Cleanroom wipers. 6. Close the right door.	 (1)  (3)
Clean the capping tray	Latex powder-free gloves Cleanroom wiper Flushing fluid	1. To reveal the capping tray, on the LCD, tap “Maintenance > Cap Maintain On”. 2. Power off the printer. 3. Open the right door. 4. Clean the capping tray with the Cleanroom wipers soaked with the flushing fluid. 5. Dry the capping tray with new and dry Cleanroom wipers. 6. Close the right door.	 (1)  (3)
Clean the baffle	Latex powder-free gloves Cleanroom wiper Industrial alcohol	1. Power off the printer. 2. Open the right door. 3. Use Cleanroom wipers soaked with industrial alcohol to remove the inks and impurities on the baffle. 4. Close the right door.	 (2)  (3)
Clean the media height sensors	Latex powder-free gloves Cleanroom wiper Industrial alcohol	1. Power off the printer. 2. Open the flip-up door and the right door. 3. Remove the dust or other impurities on the sensor with Cleanroom wipers and industrial alcohol. 4. Close the doors.	 (2)  (3)  (3)  (3)

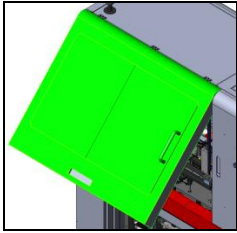
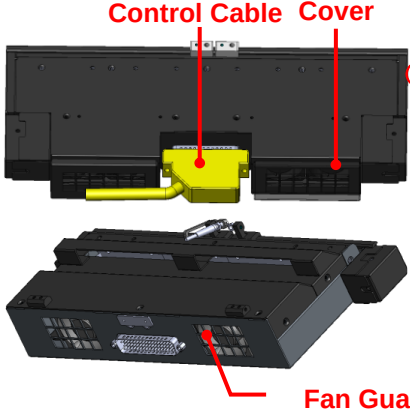


Clean the components for motion	Latex powder-free gloves Cleanroom wiper Industrial alcohol Calcium grease (AFC GRS is recommended)	1. Power off the printer. 2. Open the flip-up door. 3. Clean ① the carriage beam, ② the linear guides, ③ the jig and coupler, ④ the clamp shaft, and ⑤ the supporting wheels with Cleanroom wipers soaked with industrial alcohol.  NOTE: Apply a thin layer of calcium grease (AFC GRS is recommended) to the surface of the linear guide after cleaning it. 4. Close the flip-up door.	
Clean the doors and covers	Latex powder-free gloves Cleanroom wiper Industrial alcohol Water	1. Power off the printer. 2. Use Cleanroom wipers soaked with water or industrial alcohol to clean all the covers and doors of the printer.	
Check the filter regulator	An empty container	1. If water is in the regulator, put an empty container under the regulator. 2. Turn off the air source, and water flows out.  WARNING: Don't spill water on the live parts to avoid short circuits and electric shock. 3. Turn on the air source.	 Water flows out into the container

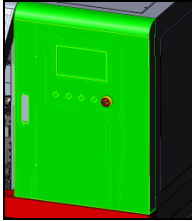
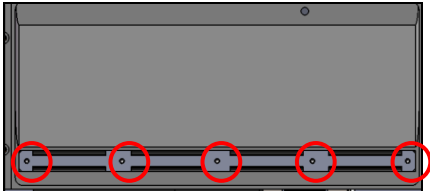
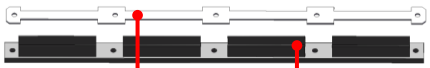
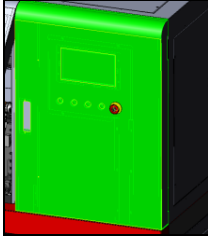
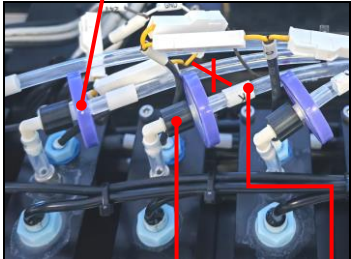
■ BIWEEKLY MAINTENANCE

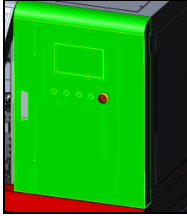
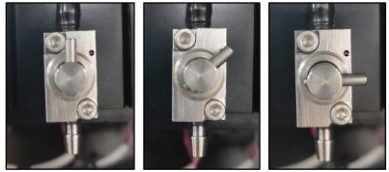
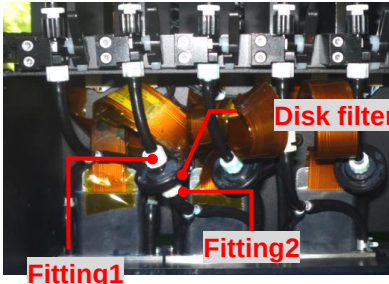
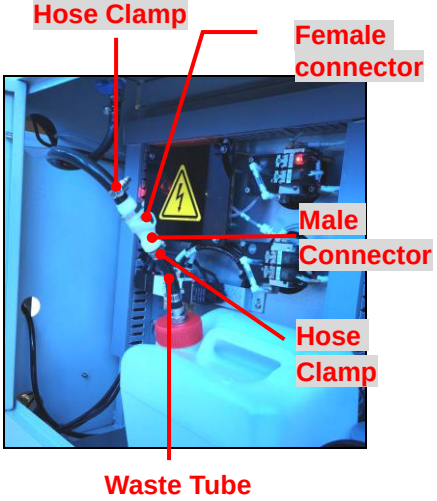
Items	Tools	Procedures	Pictures
Lubricate the lead screw	Lithium grease #2 Latex powder-free gloves	1. Open the front cabinet door. 2. Move the spin tool module up and down, until you reach the lead screw easily. 3. Add the grease directly to the lead screw. 4. Move the spin tool module up and down to evenly distribute the grease. 5. Close the cabinet door.	 (1)  Lead Screw
Checking the ink tubes and motor cables on the carriage	Latex powder-free gloves Hex key wrench set	1. Power off the printer. 2. Open the right door and uninstall the carriage cover. 3. Check the cables and the ink tubes. Make sure they do not intertwine with each other or wear out. 4. Close the door and carriage cover.	 (2)  (3)  

■ MONTHLY MAINTENANCE

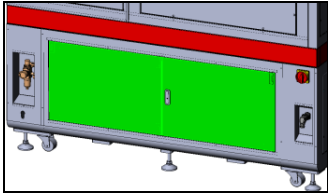
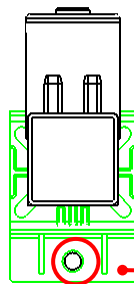
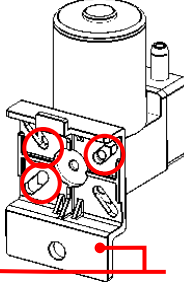
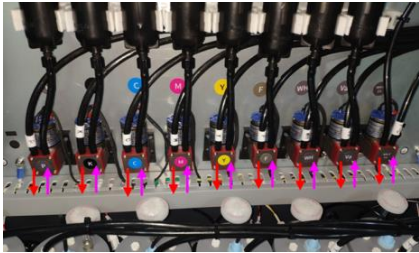
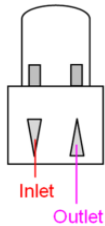
Items	Tools	Procedures	Pictures
Replace the sponge sheet of the UV lamp (ECLE400122)	Sponge sheet Cleanroom wiper Latex powder-free gloves Industrial alcohol	1. Power off the printer. 2. Open the flip-up door. 3. Disconnect the control cable. 4. Open the cover, and take out the sponge sheet. 5. Use Cleanroom wipers soaked with industrial alcohol to wipe the cover and the fan guard. ⚠ WARNING: The Cleanroom wipers cannot be too wet to prevent industrial alcohol from dripping onto the live parts. 6. Secure the new sponge sheet in the cover. 7. Close the flip-up door.	 (2)  Control Cable Cover (2) (3) Fan Guard

■ QUARTERLY MAINTENANCE

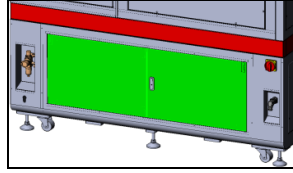
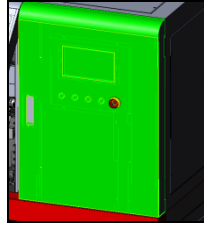
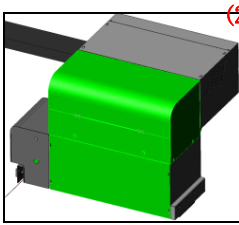
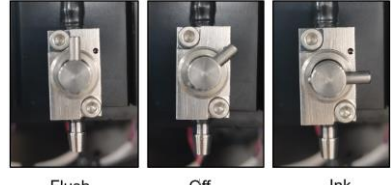
Items	Tools	Procedures	Pictures
Replace the wiping blades (MCAS000215)	Wiping blades Cleanroom wiper Latex powder-free gloves Flushing fluid Phillips screwdriver	- To reveal the wiping blades, tap the “Left Arrow” button on the touch-screen LCD. - Power off the printer. - Open the right door. - Remove the screws fixing the wiping blades. - Separate the wiping blades from the sheet metal. - Clean the new wiping blades with Cleanroom wipers soaked with the flushing fluid. - Align the wiping blades with the gaps on the sheet metal, and then fix them to the PWC with screws. - Close the right door.	 (3)  (4)  (5) Sheet metal Wiping blades
Replace the ink overflow filter (ECIK000155)	Ink overflow filter Diagonal cutting pliers Hex key wrench set	- Power off the printer. - Open the front door and uninstall the carriage covers. - Remove the current filter by loosening the Luer fitting and cutting the air tube. Cut the tube as close to the filter as possible. - Connect the new filter with the inlet to the Luer fitting and outlet to the air tube. ⚠ WARNING: Connect the Luer fitting and air tube tightly to avoid air leaks. - Install back the carriage covers.	 (2)  (3) Ink overflow filter Luer fitting Air tube Outlet Inlet with thread

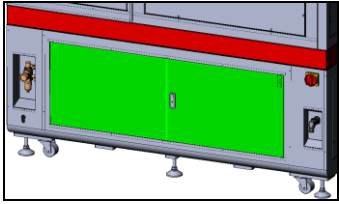
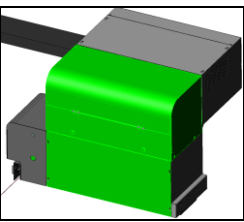
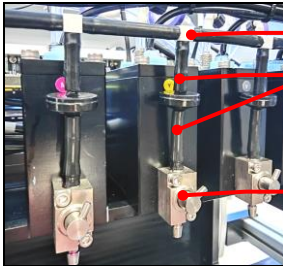
Items	Tools	Procedures	Pictures
Replace the disk filter (ECIK000051)	Disk filter Syringe Flushing fluid Cleanroom wiper Latex powder-free gloves Industrial alcohol Hex key wrench set	- Clean the new filter by purging the flushing fluid into its inlet (marked "IN") with a clean syringe. Then, empty the syringe and purge out the flushing fluid in the filter. - Power off the printer. - Open the front door and uninstall the carriage covers. - Switch the manual 3-way valve to position "Off". - Separate the disk filter with the two Luer fittings. - Connect the new filter's inlet to Luer fitting 1 and outlet to Luer fitting 2. - Remove the waste ink on the other parts with Cleanroom wipers soaked with industrial alcohol. - Switch the manual 3-way valve to position "Ink". - Close the front door and install the carriage covers. - Power on the printer. - Purge the ink so that the flushing fluid in the filter is purged out of the filter.	 (1)  (3)  (4)  (5) Labels in (5): Fitting1, Fitting2, Disk filter
Replace the quick connectors on the waste tube (ECCO001031 for male connector; ECCO001089 for female connector)	Quick connectors Cleanroom wiper Latex powder-free gloves Industrial alcohol	⚠ WARNING: Don't spill waste ink on the live parts around the waste tank. Spread several pieces of Cleanroom wipers around these areas to avoid messes. - Power off the printer. - In the front cabinet, find the waste connectors. - Press the silver thumb latch to separate the two connectors. - Loosen the hose clamps. - Disconnect the two connectors. - Plug the new connectors into the waste tubes, and fasten them with hose clamps tightly. - Plug the two connectors together. - Remove the waste ink with Cleanroom wipers soaked with industrial alcohol.	

■ SEMI-ANNUALLY MAINTENANCE

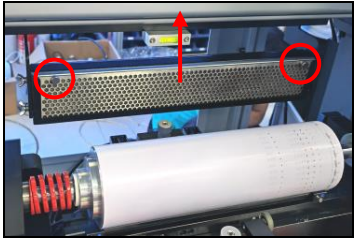
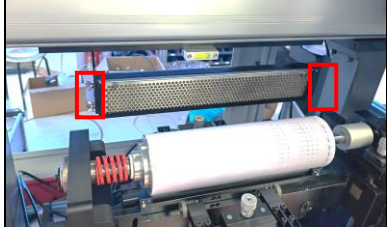
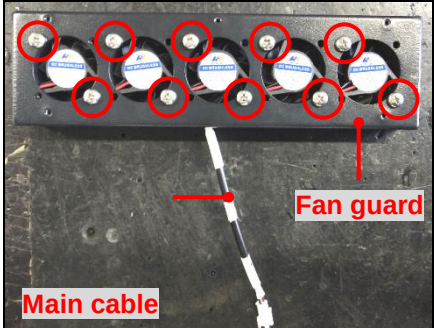
Items	Tools	Procedures	Pictures
Replace the ink pump (ECIK000024-A)	Ink pump Diagonal cutting pliers Cleanroom wiper Latex powder-free gloves Industrial alcohol Phillips screwdriver	⚠ WARNING: Don't spill ink on the live parts around the ink pumps. You can spread several pieces of Cleanroom wipers around these areas to avoid messes. - Power off the printer. - Open the rear cabinet door. - Cut and disconnect the ink tubes from the pump inlet and outlet, and then immediately block the tubes to prevent the ink from flowing out. Cut the tubes as close to the pump as possible. - Open the cable trunking and disconnect the pump cable. - Uninstall the ink pump with its holder by removing the cross recess screw. - Loosen the screws at the back of the ink pump and separate the pump from its holder. - Reverse steps 6-3 to install the new pump. ⚠ WARNING: - Connect the pump cable and the ink tubes to the corresponding pump according to the labels. - Make sure that the tubes are plugged tightly with the pump to prevent ink from flowing out. - Do not misconnect the tubes to the pump inlet and outlet: - For the ink pump KCMYWVP, the ink tube from the main ink tank goes to the pump inlet. The ink tube from the needle filter goes to the pump outlet. - For the F-Cir pump, the ink tube from the main ink tank goes to the pump outlet. The other tube goes to the pump inlet.	 (2)  (3)  (4)  (5)  (6) Pump holder  (7)  Inlet Outlet

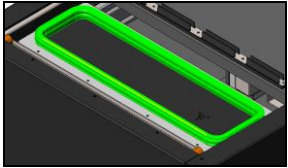
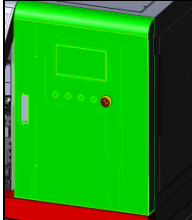
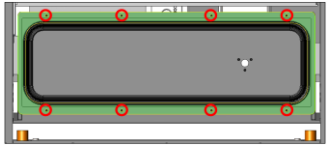


Items	Tools	Procedures	Pictures
Replace the needle filter (ECIK000254)	Needle filter Cleanroom wiper Latex powder-free gloves Industrial alcohol	⚠ WARNING: Take care not to spill ink on the live parts around the filter. You can spread several pieces of Cleanroom wipers around these areas to avoid messes. - Power off the printer. - Open the rear cabinet. - Disconnect Luer fitting 1 from the filter inlet, then block Luer fitting 1 and the filter inlet. - Disconnect Luer fitting 2 from the filter outlet, then block Luer fitting 2 and the filter inlet. - Remove the filter from the clamp. - Fix the new filter to the clamp with the arrow pointing upward. - Connect the filter inlet to Luer fitting 1 and the filter outlet to Luer fitting 2. - Remove the ink on the other parts with Cleanroom wipers soaked with industrial alcohol.	 (2)  (3)  (6)
Replace the manual 3-way valve (ECCO001055)	Manual 3-way valve Hex key wrench set Cleanroom wiper Latex powder-free gloves Diagonal cutting pliers	- Power off the printer. - Open the right door and uninstall the carriage covers. - Switch the manual 3-way to the position "Off". - Disconnect the ink tubes from the valve's inlet and outlet. Immediately block the valve's inlet and outlet. - Remove the two screws fixing the valve and then take it away. - Fix the new valve with the screw, and connect the ink tubes to the valve's inlet and outlet. ⚠ WARNING: Connect the tubes according to the labels on the ink tube assemblies. Do not misconnect them. - Wipe off the ink on the other parts, if needed, with the Cleanroom wiper soaked with industrial alcohol. - Install back the carriage covers and close the front door.	 (2)  (2)  (3)  (4)  (5)

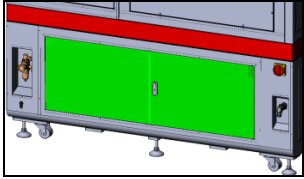
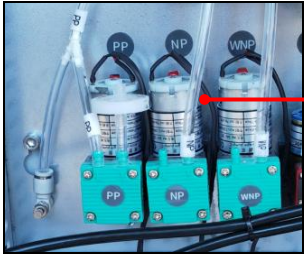
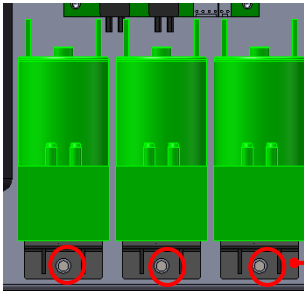
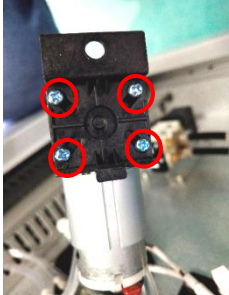
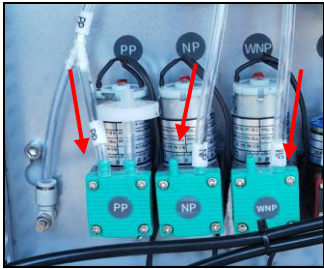
Items	Tools	Procedures	Pictures
Replace the one-way valve in the flush channel (ECCO000283)	One-way valve (black disk) Cleanroom wiper Latex powder-free gloves Diagonal cutting pliers	One-way valve above the needle filter - Power off the printer. - Open the rear cabinet. - Cut and disconnect the flush tube at the two ends of the valve. Cut the tube as close to the valve as possible. - Connect the new valve to the tube with the triangle pointing upwards. ⚠ WARNING: Make sure that the tube is plugged tightly to prevent flushing fluid from leaking. - Remove the flush on the other parts with the Cleanroom wipers. - Close the rear cabinet.	 (2)
		One-way valve above the manual 3-way valve - Power off the printer. - Open the front door and uninstall the carriage covers. - Cut the ink tubes from the position marked in the picture. - Cut two sections of short tubes that are the same length as the original ones. Connect the new valve and the short tubes to the manual 3-way valve and the white connector. 📝 NOTE: Make sure the triangle on the valve points downward.	 (3)  (4)   (2)   Connector Short Tube Manual 3-way Valve



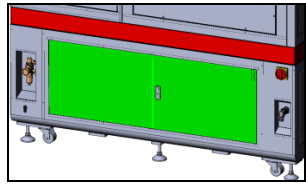
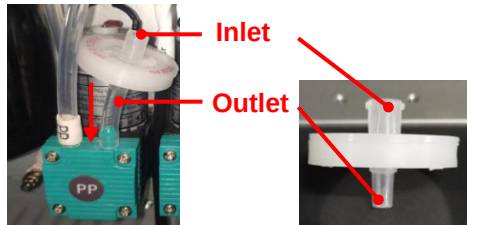
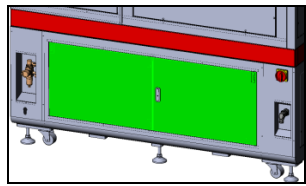
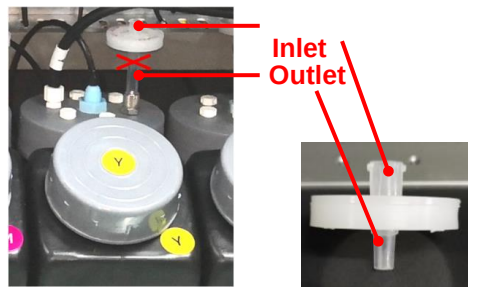
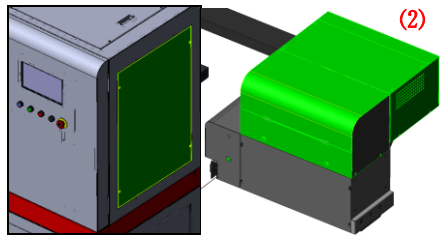
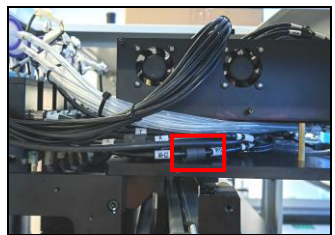
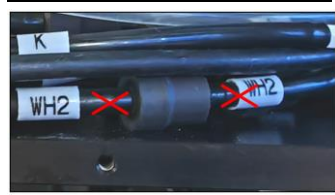
Items	Tools	Procedures	Pictures
Replace the ink mist fan (ECFN000106-A)	Ink mist fan Hex key wrench set Cleanroom wiper Latex powder-free gloves Industrial alcohol	1. Power off the printer. 2. Open the flip-up door. 3. Hold the knob and pull out the sponge sheet box. 4. Disconnect the ink mist fan cable (main cable). 5. Loosen the four wing screws to uninstall the ink mist fan module. 6. Remove the fan cover by loosening the three screws. 7. Loosen the screws fixing each fan to the fan guard, disconnect the cable of each fan with the main cable, and then take away the fans. 8. Clean the fan guard, fan cover, and sponge sheet box with the Cleanroom wipers soaked with industrial alcohol. ⚠ WARNING: The Cleanroom wiper cannot be too wet to prevent the industrial alcohol from dripping onto the live parts. 9. Install the new fans to the fan guard and connect their cables to the main cable. Arrange the cable in the fan guard. 10. Install back the fan cover. 11. Install back the ink mist fan module with four wing screws and connect the main cable back. 12. Put back the sponge sheet box. 13. Adjust the position of the ink mist collector. See the printer's <i>User Manual</i>. 14. Close the flip-up door.	 (3)  (4) Ink Mist Fan Cable (Main Cable)  (5)  (6)  (7) Fan guard Main cable

Items	Tools	Procedures	Pictures
Replacing the sealing rubber strip (MCOT000516)	Sealing rubber strip Hex key wrench set Cleanroom wiper Latex powder-free gloves Industrial alcohol Cyanoacrylate adhesive	- To reveal the sealing rubber strip, tap “Maintenance > Cap Maintain On” on the touch-screen LCD. - Power off the printer. - Open the right door. - Remove the eight screws, take away the mounting plate, and then remove the old strip. - Clean the capping tray, the mounting plate, and the new sealing rubber strip with the Cleanroom wiper soaked with flushing fluid. - Press the new sealing rubber strip into the mounting plate, starting from the middle position (marked with a red line) with the soft side on top. - Glue the two ends with the cyanoacrylate adhesive. Do not apply too much cyanoacrylate adhesive. - Install back the mounting plate with the eight screws. - Close the right door.	 (1)  (3)  (4)  (6) (7)  (6) Soft Side with Strips

■ ANNUALLY MAINTENANCE

Items	Tools	Procedures	Pictures
Replace the air pump (ECIK000026-A)	Air pump Diagonal cutting pliers Phillips screwdriver	- Power off the printer. - Open the rear cabinet. - Disconnect the air pump cables. - Cut the air tubes connected to the pumps and as close to the pump as possible. And then disconnect the tubes from the pumps. Also, cut the tube at the air disk filter end and as close to the filter as possible. And then disconnect the tube. Keep the tube to install the new air disk filter in “Replace the air disk filter”. - Loosen the screw and uninstall the pump together with the pump holder. - Uninstall the air pump from its holder by removing the four screws. - Fix the new air pump to the holder, and then fix the holder to the printer. - Connect the air pump cables according to the labels (NP, WNP, PP) on the cables. ⚠ WARNING: Do not misconnect the cables. - Connect the air tubes to the corresponding air pumps according to the labels (NP, WNP, PP). ⚠ WARNING: Do not misconnect the tubes, and make sure that all the tubes are plugged tightly to the pump to avoid air leaks. - Connect the disk filter to the PP pump. See “Replace the air disk filter”.	 (2)  (3) Pump cables  (4) Air filter  (5) Pump holder  (6)  (9)



Items	Tools	Procedures	Pictures
Replace the air disk filter (ECIK000021)	Air disk filter Diagonal cutting pliers	Filter on the PP pump - In the rear cabinet, find the PP pump. - Prepare the tube disconnected from the PP pump in step 4 in "Replace the air pump". - Connect the outlet of the filter to the air tube. Connect the other end of the tube to the PP pump. ⚠ WARNING: Make sure the tube is plugged tightly to prevent air leaks. Filter on the main ink tank - Find the main ink tanks in the rear cabinet. - Cut the air tube as close to the filter as possible. And then remove the filter. - Connect the outlet of the new filter to the air tube. ⚠ WARNING: Make sure the tube is plugged tightly to prevent air leaks.	   
Replace the one-way valve in the ink channel (ECCO001029-A)	One-way valve Hex key wrench set Cleanroom wiper Latex powder-free gloves Industrial alcohol Diagonal cutting pliers	- Power off the printer. - Open the right cover and the carriage covers. - Put Cleanroom wipers under the current one-way valves. - Cut the tubes at the valve's two ends and as close to the valve as possible, and then remove the valve. - Plug the new valve into the ink tube, with the arrow pointing to the sub-ink-tank. You can feel the arrow by pressing it. - Wipe off the inks leaked with a Cleanroom wiper and industrial alcohol. - Arrange the ink tubes. - Install back the carriage's covers and close the right cover. - Install back the covers.	    

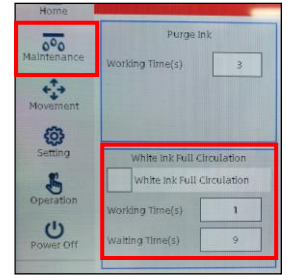


Items	Tools	Procedures	Pictures
Replace the ink tube assembly (MBIA000074-A; MBIA000073-A)		Contact your technical support	



■ MAINTENANCE WHILE THE PRINTER IS IDLE

⚠ WARNING: Always keep the “White Ink Full Circulation” check box selected in the “Maintenance” menu, to avoid white ink sedimentation while the printer is idle. By default, the check box is selected.



Idle Time	Tools	Maintenance	Operation
Over 30 minutes (e.g. lunch break)	/	Cap the print heads with the capping tray.	Before idle: PCS orders the printer to cap the print heads automatically. Before using the printer: 1. Uncap the print heads. 2. Perform an auto clean. 3. Print check nozzle test chart to inspect head nozzle condition. If nozzles are jammed, immediately purge the ink or flushing fluid to clear the nozzles.
Less than 2 days (e.g. after the workday or at the weekend)	Cleanroom wiper Latex powder-free gloves Flushing fluid	Power on and power off the printer.	Before idle: Power off the printer according to the “Daily Power-off Procedures”. Before using the printer: 1. Perform steps 1-5 in the “Daily Power-on Procedures”. 2. Wait one minute for the white ink circulation, then perform an auto clean. NOTE: Because the white ink is easy to sediment, power on the printer 20-30 minutes before printing for the white ink circulation and to avoid sedimentation. 3. Print check nozzle test chart to inspect the head nozzle condition. If nozzles are jammed, immediately purge the ink or flushing fluid to clear the nozzles. 4. Inspect the software settings of PCS and the ColorPRINT Bottle. Make sure all parameters are correctly set.
3-7 days (e.g. long holidays)	Cleanroom wiper Latex powder-free gloves Flushing fluid Hex key wrench set Nose pliers	Clean the white print head and power on or power off the printer.	Before idle: 1. Perform steps 1-3 in the “Daily Power-off Procedures”. 2. Purge out the ink in the white head with the flushing fluid according to “White Head Cleaning Procedures”. 3. Perform step 4 in the “Daily Power-off Procedures”. Before using the printer: 1. Perform the steps in the “Ink Filling Procedure after Cleaning Print Heads” to purge out the flushing fluid in the print heads with the ink. 2. Perform steps 6-8 in the “Daily Power-on Procedures”.
More than 7 days	Cleanroom wiper Latex powder-free gloves Flushing fluid Hex key wrench set Nose pliers	Clean all print heads and power on or power off the printer.	Before idle: 1. Perform steps 1-3 in the “Daily Power-off Procedures”. 2. Purge out the ink in the print heads with the flushing fluid according to “White Head Cleaning Procedures” and “Color Head Cleaning Procedures”. 3. Perform step 4 in the “Daily Power-off Procedures”. Before using the printer: 1. Perform the steps in the “Ink Filling Procedure after Cleaning Print Heads” to purge out the flushing fluid in the print heads with the ink. 2. Perform steps 6-8 in the “Daily Power-on Procedures”.



Idle Time	Tools	Maintenance	Operation
Unknown	/	Power on the printer and maintain the heads.	Power on the printer according to the “<u>Daily Power-on Procedures</u>” and perform the maintenance operations. ● In the PCS, click “Head Maintain” in the “Setting” tab: ▪ Select the “Spit” check box under the “During Idle” pane and set the interval to less than 10 seconds. ▪ Set the “Idle Time before Auto Capping” to 30 minutes. ● Every two hours: 1) Uncap the print heads. 2) Perform an auto clean. 3) Print check nozzle test chart to inspect the head nozzle condition. If nozzles are jammed, immediately purge the ink or flushing fluid to clear the nozzles.

Daily Power-on Procedures	Daily Power-off Procedures
1. Turn on the computer installed with PCS and ColorPRINT Bottle software. 2. Start the PCS and the ColorPRINT Bottle. 3. Press the “ON” button to power on the printer. The printer initializes after powering on. After the initialization, the LCD displays “Ready”. 4. Check whether any warning message appears on the LCD and PCS. If any warning message appears, solve the problem according to the hint; or on the PCS, click “Help” behind the warning. 5. Switch the manual 3-way valve to position “Ink”. 6. Perform an auto clean with the LCD or PCS. 7. Print the check nozzle test chart to inspect the head nozzle condition. If nozzles are jammed, immediately purge the ink or flushing fluid to clear the nozzles. Write down the date, time, and your name on each test chart and save it for future checking. 8. Inspect the software settings of PCS and the ColorPRINT Bottle. Make sure all parameters are correctly set.	1. Switch the manual 3-way valve to position “Ink”. 2. Perform an auto clean with the LCD or PCS. 3. Print the check nozzle test chart to inspect the head nozzle condition. If nozzles are jammed, immediately purge the ink or flushing fluid to clear the nozzles. Write down the date, time, and your name on each test chart and save it for future checking. 4. Maintain the heads and power off the printer. With the “OFF” button and the LCD: 1) Press the “OFF” button for less than three seconds. NOTE: If pressing the “OFF” button for more than three seconds, the printer directly powers off. 2) Select “After maintain head” and then press the “Enter” key. The printer purges the ink, wipes the nozzles, and caps the print heads. 3) Switch the manual 3-way valve to position “OFF”. 4) Select “OK” and press the “Enter” key to power off the printer. With the PCS: 1) On the “Operation” tab of the PCS, select “Maintain Head & Power off”. NOTE: If you click “Power Off”, the printer powers off directly. 2) Click “OK”. The printer purges the ink, wipes the nozzles, caps the print heads, and then powers off. 3) Switch the manual 3-way valve to position “OFF”.

**Color Head Cleaning Procedures**

1. Switch the manual 3-way valve to the position "Flush".
2. In the PCS, press and hold the flush button (F) for **10 seconds**.
3. Wait for **10 seconds** and repeat step 2.
Repeat steps 2 and 3 for **3 times**.
4. Wipe the print heads.
5. Clean the head nozzle surface, the head plate, and the wiping blades with new Cleanroom wipers.
6. In the PCS, press and hold the flush button (F) for **20 seconds**.
7. Wipe the print heads.
8. Perform steps 6, 7, and 5 in turn, and repeat **4 times**.
9. Switch the manual 3-way valve to position "OFF".
10. Disconnect the Luer fitting from the outlet of the disk filter, cap the ink tube, and then uninstall the print head. Wrap the print head with new Cleanroom wipers and the cling film.

 **WARNING:** Contact your technical support if you have any doubts about uninstalling the print head to avoid any damage to the head.

11. Block the outlet of the disk filter with the Luer cap.

White Head Cleaning Procedures

1. Switch the manual 3-way valve to the position "Flush".
2. In the PCS, press and hold the flush button (F) for **10 seconds**.
3. Wait for **10 seconds** and repeat step 2.
Repeat steps 2 and 3 for **5 times**.
4. Wipe the print heads.
5. Clean the head nozzle surface, the head plate, and the wiping blades with new Cleanroom wipers.
6. In the PCS, press and hold the flush button (F) for **25 seconds**.
7. Wipe the print heads.
8. Perform steps 6, 7, and 5 in turn, and repeat **8 times**.
9. Switch the manual 3-way valve to position "OFF".
10. Disconnect the Luer fitting from the outlet of the disk filter, cap the ink tube, and then uninstall the print head. Wrap the print head with new Cleanroom wipers and the cling film.

 **WARNING:** Contact your technical support if you have any doubts about uninstalling the print head to avoid any damage to the head.

11. Block the outlet of the disk filter with the Luer cap.

Ink Filling Procedures after Cleaning Print Heads

1. Install the print head and connect the ink tube to the disk filter.

 **WARNING:** Contact your technical support if you have any doubts about installing the print head to avoid any damage to the head.

2. Power on the printer.
3. Start the PCS.
4. Switch the manual 3-way valve to position "Ink".
5. Purge the ink with the PCS or the LCD.
6. Wipe the print head.
7. Repeat steps 5 and 6 for **10 times**.
8. Perform an auto clean.
9. Print the check nozzle test chart.
10. Check whether the test chart is cured completely.
If the test chart is not cured completely due to flush residual in the print head, repeat steps 5 and 6.

■ MAINTENANCE CHECKLIST

WEEKLY MAINTENANCE CHECKLIST

Interval	Items	Completed by	Completed date
Daily	Follow the procedure to: ☐ clean the head plate and the PWC ☐ inspect the head nozzles		
Daily	Follow the procedure to: ☐ clean the head plate and the PWC ☐ inspect the head nozzles		
Daily	Follow the procedure to: ☐ clean the head plate and the PWC ☐ inspect the head nozzles		
Daily	Follow the procedure to: ☐ clean the head plate and the PWC ☐ inspect the head nozzles		
Daily	Follow the procedure to: ☐ clean the head plate and the PWC ☐ inspect the head nozzles		
Daily	Follow the procedure to: ☐ clean the head plate and the PWC ☐ inspect the head nozzles		
Daily	Follow the procedure to: ☐ clean the head plate and the PWC ☐ inspect the head nozzles		
Every two days	CLEAN the following parts: ☐ UV emitting window ☐ UV light sensor ☐ UV light shield sensor		
Every two days	CLEAN the following parts: ☐ UV emitting window ☐ UV light sensor ☐ UV light shield sensor		
Every two days	CLEAN the following parts: ☐ UV emitting window ☐ UV light sensor ☐ UV light shield sensor		
Every five days	CLEAN the following parts: ☐ ink mesh		



MONTHLY MAINTENANCE CHECKLIST

Interval	Items	Completed by	Completed date
Week 1	CLEAN the following parts: ☐wiping tray ☐capping tray ☐baffle ☐media height sensors ☐printer movement parts ☐printer doors and covers CHECK the following parts: ☐filter regulator		
Week 2	CLEAN the following parts: ☐wiping tray ☐capping tray ☐baffle ☐media height sensors ☐printer movement parts ☐printer doors and covers CHECK the following parts: ☐filter regulator ☐ink tubes and motor cables on the carriage LUBRICATE the following parts: ☐lead screw for the spin tool module		
Week 3	CLEAN the following parts: ☐wiping tray ☐capping tray ☐baffle ☐media height sensors ☐printer movement parts ☐printer doors and covers CHECK the following parts: ☐filter regulator		
Week 4	CLEAN the following parts: ☐wiping tray ☐capping tray ☐baffle ☐media height sensors ☐printer movement parts ☐printer doors and covers CHECK the following parts: ☐filter regulator ☐ink tubes and motor cables on the carriage LUBRICATE the following parts: ☐lead screw for the spin tool module		



ANNUAL MAINTENANCE CHECKLIST

Interval	Items	Completed by	Completed date
Month 1	REPLACE the following parts: ☐ sponge sheet of UV lamp		
Month 2	REPLACE the following parts: ☐ sponge sheet of UV lamp		
Month 3	REPLACE the following parts: ☐ sponge sheet of UV lamp ☐ wiping blades ☐ disk filter ☐ quick connectors on the waste tube ☐ ink overflow filter		
Month 4	REPLACE the following parts: ☐ sponge sheet of UV lamp		
Month 5	REPLACE the following parts: ☐ sponge sheet of UV lamp		
Month 6	REPLACE the following parts: ☐ sponge sheet of UV lamp ☐ wiping blades ☐ disk filter ☐ ink overflow filter ☐ ink mist fan ☐ ink pump ☐ needle filter ☐ manual 3-way valve ☐ sealing rubber strip ☐ quick connectors on waste tube ☐ one-way valve in flush channel		
Month 7	REPLACE the following parts: ☐ sponge sheet of UV lamp		
Month 8	REPLACE the following parts: ☐ sponge sheet of UV lamp		
Month 9	REPLACE the following parts: ☐ sponge sheet of UV lamp ☐ wiping blades ☐ disk filter ☐ quick connectors on the waste tank ☐ ink overflow filter		
Month 10	REPLACE the following parts: ☐ sponge sheet of UV lamp		
Month 11	REPLACE the following parts: ☐ sponge sheet of UV lamp		
Month 12	REPLACE the following parts: ☐ sponge sheet of UV lamp ☐ wiping blades ☐ disk filter ☐ ink overflow filter ☐ ink pump ☐ needle filter ☐ manual 3-way valve ☐ ink mist fan ☐ air disk filter ☐ air pump ☐ sealing rubber strip ☐ quick connectors on waste tube ☐ one-way valve in flush channel ☐ one-way valve in ink channel ☐ ink tube assembly (Contact technical support)		